

9-Button Priority Keypad Project

Breadboard Assembly Guide

Component: 74LS147 & Arduino Uno

December 5, 2025

1 Project Overview

This project builds a decimal input keypad (1-9) using the ****74LS147 Priority Encoder****. This chip takes 9 input lines and converts them into a 4-bit Binary Coded Decimal (BCD) output.

Key Logic Concepts:

- **Priority Encoding:** If multiple keys are pressed, only the highest number is output.
- **Active Low Inputs:** The chip activates when inputs are pulled to **Ground (LOW)**, not 5V.
- **Inverted Outputs:** The output is the inverse of the binary number (e.g., 9 is normally 1001, but this chip outputs 0110). We will handle this in wiring/code.

2 Bill of Materials Checklist

Component	Quantity/Notes
74LS147	10-to-4 Priority Encoder
Pushbuttons	9x (Tactile Switch)
Resistors (Pull-Up)	9x 10k Ω (or 4.7k Ω for LS logic strong pull-up)
LEDs	4x (Red/Green/Blue)
Resistors (Current Limit)	4x 220 Ω
Arduino Uno	For reading data
Jumper Wires	Male-to-Male
Breadboard	Standard size
Optional (Debounce)	74HC14, 10k Ω , 0.1 μ F

3 Phase 1: Breadboard Layout & Wiring

3.1 Step 1: Power Distribution

1. Connect the Arduino **5V** pin to the breadboard **Red Rail (+)**.
2. Connect the Arduino **GND** pin to the breadboard **Blue Rail (-)**.
3. Place the **74LS147** chip in the center of the breadboard.
4. **Power the Chip:**
 - Pin 16 (VCC) $\rightarrow$ Red Rail (+5V)
 - Pin 8 (GND) $\rightarrow$ Blue Rail (GND)

3.2 Step 2: The Buttons (Input Logic)

Since the 74LS147 uses **Active Low** inputs, the buttons must connect the pin to Ground when pressed. The resistors must pull the pin to 5V when not pressed.

For each Button (1 through 9):

1. Connect one leg of the pushbutton to the **Ground Rail**.
2. Connect the other leg of the pushbutton to the corresponding **74LS147 Input Pin** (see table below).
3. Connect a **10k Ω (or 4.7k Ω)** resistor from the **Same Input Pin** to the **5V Rail**.

Button Number	74LS147 Pin	Description
1	11	Input 1
2	12	Input 2
3	13	Input 3
4	1	Input 4
5	2	Input 5
6	3	Input 6
7	4	Input 7
8	5	Input 8
9	10	Input 9
(NC)	15	Not Connected

Table 1: Input Pin Mapping

3.3 Step 3: Visual Output (LEDs)

The outputs are inverted. To make the LEDs light up for a logical "1" when the output is actually Low (0V), we wire them in "Active Low Source" mode.

Wiring 4 LEDs (A, B, C, D):

- **Anode (Long Leg):** Connect directly to **5V Rail**.
- **Cathode (Short Leg):** Connect to a 220 Ω resistor.
- **Resistor Other End:** Connect to the **74LS147 Output Pin**.

BCD Bit	74LS147 Pin	Arduino Connection
A (LSB)	9	Digital Pin 2
B	7	Digital Pin 3
C	6	Digital Pin 4
D (MSB)	14	Digital Pin 5

Table 2: Output Pin Mapping

4 Phase 2: Hardware Debouncing (Optional)

You mentioned having a 74HC14 (Schmitt Trigger), 10k resistor, and 0.1 μ F capacitor. This creates a hardware debouncer for **one single button**. To debounce the entire keypad in hardware, you would need 9 sets of these.

Wiring for Single Button Debounce:

1. **Switch:** Connect Button leg to Ground.
2. **Pull-Up:** Connect 10k Ω resistor from Button leg to 5V.
3. **RC Filter:**
 - Connect a 4.7k Ω (or 10k) resistor from the Button/Pull-Up junction to the 74HC14 Input (e.g., Pin 1).
 - Connect the 0.1 μ F capacitor from 74HC14 Input (Pin 1) to Ground.
4. **Inverter:** The 74HC14 Output (Pin 2) goes to the 74LS147 Input.

Note: For this project, software debouncing in Arduino is easier than building 9 of these circuits.

5 Phase 3: Arduino Code

This code reads the inverted signals and prints the actual pressed number to the Serial Monitor.

```
1 // 74LS147 to Arduino Interface
2 // Inputs are connected to Pins 2, 3, 4, 5
3
4 const int pinA = 2; // 74LS147 Pin 9
5 const int pinB = 3; // 74LS147 Pin 7
6 const int pinC = 4; // 74LS147 Pin 6
7 const int pinD = 5; // 74LS147 Pin 14
8
9 void setup() {
10   Serial.begin(9600);
11   pinMode(pinA, INPUT);
12   pinMode(pinB, INPUT);
13   pinMode(pinC, INPUT);
14   pinMode(pinD, INPUT);
15   Serial.println("74LS147 Keypad Ready");
16 }
17
18 void loop() {
19   // Read the raw states
20   int a = digitalRead(pinA);
21   int b = digitalRead(pinB);
22   int c = digitalRead(pinC);
23   int d = digitalRead(pinD);
24
25   // 74LS147 Outputs are Active LOW (Inverted)
26   // Logic 0 (0V) means the bit is Active (1)
27   // Logic 1 (5V) means the bit is Inactive (0)
28
29   // Invert reads to get standard binary
30   int valA = !a;
31   int valB = !b;
32   int valC = !c;
33   int valD = !d;
34
35   // Calculate Decimal Value
36   // D is 8, C is 4, B is 2, A is 1
37   int decimalValue = (valD * 8) + (valC * 4) + (valB * 2) + (valA * 1);
38
39   // 0 means no key pressed (or technically 0, but 74147 is 1-9)
40   // Actually, 74LS147 outputs 1111 (High) when no key is pressed.
41   // Our inversion (!) makes that 0000.
```

```
42
43 if (decimalValue > 0) {
44     Serial.print("Key Pressed: ");
45     Serial.println(decimalValue);
46
47     // Simple software debounce delay
48     delay(200);
49 }
50 }
```

6 Troubleshooting

1. **All LEDs are ON:** Check if you wired the LEDs to Ground instead of 5V. The chip outputs LOW for active bits. If wired to ground, they will only light up when the bit is "Off" (High).
2. **Garbage Output:** 74LS inputs float HIGH, but they are sensitive. Ensure all 9 Pull-Up resistors are firmly connected to 5V.
3. **Number 9 doesn't work:** Check Pin 10 of the chip. It is easily confused with Pin 9 (Output A).